

Camera-Based Vision System for a Formula Student Driverless Vehicle

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Abstract

Accurate real-time cone detection is essential for Formula Student (FS) driverless vehicles, which operate at high speeds. Many FS teams rely on LiDAR for perception, viewing cameras as a cheaper but less accurate alternative. Advances in neural networks have made camera-based systems more competitive. Monash Motorsport currently uses LiDAR, incurring penalties in weight, cost, and power consumption. A camera-based system could reduce these penalties while maintaining high detection accuracy and low latency. Developing such a system involves collecting and annotating data, training a lightweight model, and testing both offline and on-track. Initial choices include a Basler acA2440-20gc camera, Grounding DINO for auto-labelling, and a YOLO11n model for on-track use. This research explores a camera-based perception stack for Monash Motorsport, providing insights into its performance relative to LiDAR and informing future vehicle design decisions.

1 Introduction

Within the wider Formula Student competition, Formula Student Driverless represents the autonomous category, where student teams develop and build a single-seater, open-wheel race car capable of navigating a track without human control. To achieve this, driverless vehicles rely on vision systems to act as their eyes, enabling them to perceive their surroundings and make decisions based on visual input. As such, the performance of a Formula Student driverless vehicle can be largely attributed to the strength and quality of these vision systems. The majority of vision systems can be fitted into three categories: Light Detection And Ranging (LiDAR), camera (utilizing computer vision), and LiDAR-camera fusion, all with their own benefits and pitfalls. LiDAR has traditionally been regarded as the most accurate, but also the most expensive. In contrast, camera systems offer a much more cost-effective alternative, but historically suffer from lower perception accuracy. However, advancements in neural networks and machine learning algorithms in recent years have heavily improved camera-based perception, leading to an increase in popularity among student teams.

Monash Motorsport's (MMS) M25 vehicle currently uses a Baraja LiDAR for its perception stack. This system comes with significant penalties in weight, price and power consumption, as well as limitations with colour costing and cone detection accuracy. This project aims to explore camera-based alternatives to the current system to reduce these disadvantages without heavily compromising on object detection accuracy on track. This project will focus mainly on the software implementation of such a system, but will also include a hardware prototype to evaluate the feasibility of the system in practice.

The main objectives of this project are as follows:

- Design and prototype a camera-only vision system specific to the Formula Student Driverless problem.
- Integrate the system with the rest of the MMS autonomous pipeline.
- Test the feasibility of the system on track, in real time.
- Evaluate the system against the current LiDAR-only system (see Section 3).

Through design, prototyping, testing and evaluation, this project will assess the feasibility of a camera-based vision system and allow Monash Motorsport to make better informed decisions on future vehicle designs. On a broader scale, the outcomes of this research will contribute to the discourse regarding camera vision in autonomous vehicles, and how it compares to the more established LiDAR approach.

2 Literature Review

A research report by student team KA-RaceIng (Alvarez et al., 2022) details their autonomous pipeline that led to multiple competition wins in 2021. The report details a computer vision model, powered by three front-facing RGB mono cameras used in tandem to provide a field of view (FOV) of 180°. This system allows for the classification of specific cone types based on size and colour, something that is not achievable with the Baraja LiDAR. Additionally, the report highlights the importance of innovation in Formula Student, as the event is an engineering design competition at its core. Whilst KA-RaceIng opted out of using their three-camera system on competition day, being able to showcase creativity and evaluate multiple approaches gained them additional points during their 2021 campaign.

Another student team paper, authored by members of Chalmers Formula Student (Svecovs and Hörnschemeyer, 2020) details their experience working with a camera-based vision system. The software process involves manually annotating frames and using them to train a tiny-sized YOLOv3-based (You Only Look Once) neural network to run on track. It was noted that a full-sized YOLOv3 model was also trained, but maintained the same flaws as its smaller counterpart whilst also doubling execution time. Two approaches were used to calculate cone positions: camera depth and cone size. The Stereolabs ZED camera automatically detects the distance to the cones, whereas the size of detected cones were compared to the known sizes of the cones in Formula Student competition. Through testing, it was concluded that the camera depth method worked well at close range, but the cone size method was more accurate at longer distances. By combining the two approaches, Chalmers FS were able to develop a robust algorithm with much less error. However, this implementation puts the system at risk if Formula Student rules were to change. If the sizes of the cones used on track changed, the YOLOv3 neural network would estimate cone positions based on an incorrect assumption, leading to inaccurate results.

In terms of available models, Zhao (2023) identifies the Grounded-Segment Anything Model (Ren et al., 2024) as an all-purpose object detector suitable for autonomous driving. Grounded-SAM is a combination of Grounding DINO (Liu et al., 2023) and Segment Anything Model (Kirillov et al., 2023). First, Grounding DINO is used to take a text prompt and draw a bounding box around the object described in the text. Then, SAM is used to create an exact-pixel mask of the object within the bounding box. Despite this, the most important information to gather from each image is the center-point and colour of an object, meaning that the SAM portion of Grounded-SAM will most likely not be beneficial (masks are less suited for the Formula Student driverless problem), making Grounding DINO alone a better option.

However, Grounding DINO cannot simply be run on the vehicle in real-time. A recent study (Son and Jung, 2024) describes it as slower than YOLO models, and therefore unsuitable for real-time use cases. The study concluded that Grounding DINO is better utilized as a teacher model which can automatically label training data to train lightweight models like YOLO. Additionally, using auto-labelled data does not lead to a degradation in performance when compared to hand-labelled data. By automating the annotation process, significant amounts of time can be saved, especially when training with large amounts of data.

Past papers often evaluate the performance of a camera-based vision system using older versions of YOLO, such as v3 or v5. Whilst the older YOLO models had issues with poor object detection accuracy, a review of the YOLO algorithm in the context of autonomous driving (Wei et al., 2025) demonstrates that the detection accuracy of YOLO has improved significantly with each new version.

A paper comparing the performance of Convolutional Neural Networks (CNNs) such as YOLO, and Vision Transformers (ViTs) for the purposes of real-time autonomous driving (Gupta et al., 2025) concluded that CNNs are generally faster and more accurate, but perform worse than ViTs in challenging conditions such as low lighting and occlusions.

However, a white paper detailing the specifics of YOLO11 (Alif, 2024), the latest YOLO version released by Ultralytics, depicts it as one of the most accurate CNNs to date. With improved convolution-based attention modules, deeper feature extraction layers, and a new anchor-free approach, Alif (2024) highlights its robustness across different environmental conditions, minimizing the gap between CNNs and ViTs such as RT-DETRv2 (Lv et al., 2024) in

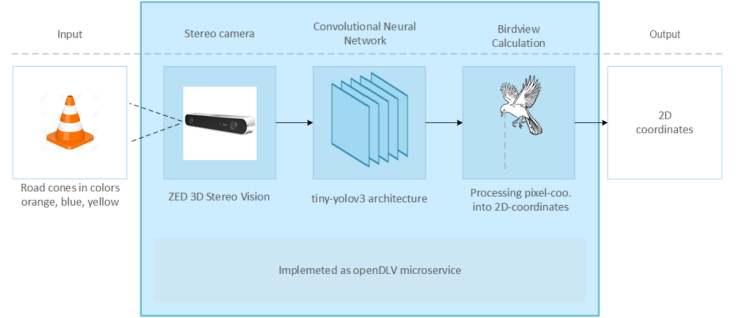


Figure 1: Chalmers FS’s camera system overview

that aspect, whilst maintaining its advantage in efficiency. While this paper evaluates YOLO11 for vehicle detection, this project focuses on cone detection. Differences in object size, shape, and appearance may lead to variations in performance metrics.

3 Methodology

The proposed camera-based vision system consists of a set of two front-facing cameras connected to an onboard computer, running a neural network-based object detection pipeline to identify cones in front of the vehicle and provide their pose to M25’s path planning section. This system should be fully integrated into the MMS autonomous pipeline.

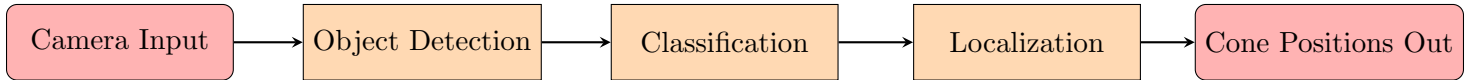


Figure 2: General camera system overview

A key challenge in this project is developing a vision algorithm capable of fully utilizing a dual-camera setup. The selected hardware for this purpose is the Basler acA2440-20gc, readily available and within MMS possession. The Basler comes with a resolution of 5 MP, a default frame rate of 22.7 FPS, and a 50° FOV. Additionally it uses a global shutter, reducing motion distortion commonly experienced with rolling shutter cameras. Initially one camera will be used, to simplify setup and evaluation efforts. If few difficulties are encountered, the system will be upgraded to include two cameras in a stereo configuration (requiring the creation of a mount). The camera(s) will be connected to an onboard NVIDIA Jetson Orin, allowing for CUDA acceleration on neural networks. If the project maintains its schedule, a mounting rig will be designed and produced in order to test the system on M25.

To gather training data, a single camera will be fixed to the M25 vehicle and recording during testing sessions, regardless of whether there is a driver inside or not. If M25 is unavailable, a tripod will be available to set up at the correct height to simulate vehicle conditions, and cones will be arranged in different configurations and with different lighting conditions to collect a wide variety of data. Once collected, video footage will be split into frames and personally selected. Chosen frames will be fed into a Grounding DINO model with prompts to identify different coloured cones. When the bounding boxes have been obtained, the original frames and the ground truths will be used to train a CNN-based model to be used on the track. The first iteration will employ a YOLO11n model for training and testing. This serves as a baseline, and the final model selection may change based on key performance metrics such as latency and detection accuracy. 80% of the data will be fed into the model for training, and 20% will be saved for testing and validation of the model. The bounding box data will be used to generate an array of cone poses, which will be sent to the MMS path planning node.

To evaluate the system’s performance, four key metrics will be tracked: detection accuracy, false positive rate, latency (ms) and floating point operations per second (FLOPS). The test environment will start with offline dataset testing, before moving onto controlled on-track testing. Finally, these metrics will be benchmarked against M25’s current LiDAR vision system. Detection accuracy and false positive rate will directly compare the capabilities of both systems, whilst latency offers insights into the system’s real-time responsiveness. Finally, FLOPS provides an unbiased measure of computational load, regardless of hardware specifics.

For software development, Python will be the initial programming language, for its compatibility with PyTorch, a library known for its strong support of custom YOLO models and simple prototyping. If necessary, a transition to C++ will be carried out in order to integrate into M25’s pre-existing pipeline. Version control and project management will be facilitated by Git and Bitbucket. Task tracking, issue management and sprint planning will be handled with Jira to ensure tasks and milestones remain on schedule.

4 Discussion

Preliminary testing with the Grounding DINO model hosted by Hugging Face shows promising results. When tested on video footage captured in Monash Motorsport’s Europe campaign in 2024, the model was able to identify most cones on each image and create accurate bounding boxes around each cone.

Issues were encountered when dealing with lower quality videos (720p, 360p). On average, only 20-25% of cones were accurately identified on 360p footage. However, it is expected that the chosen camera (Basler ace) will be able to record at 1080p consistently, given the 5 MP resolution.

Another issue encountered during preliminary testing was the lack of an available NVIDIA Graphics Processing Unit (GPU). Without an NVIDIA GPU, object detection models such as YOLO and Grounding DINO cannot utilize CUDA acceleration, meaning that the runtime of their algorithms is much slower. For this reason, it was difficult to test these models whilst working away from the M25 vehicle.

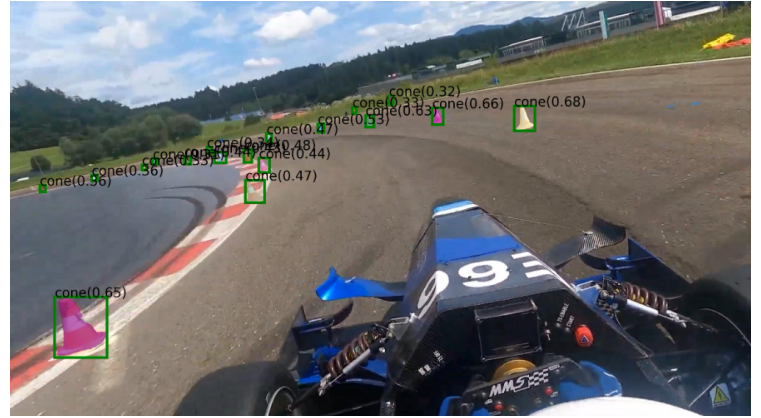


Figure 3: Grounding DINO on 1080p frame

5 Conclusion

This paper presented the concept of a camera-based vision system for a Formula Student driverless vehicle, in contrast to Monash Motorsport's current LiDAR-based system. The study conducted showcased the potential of a camera-based system, and how it can address the limitations of the LiDAR without compromising on important attributes, like detection accuracy and false positive rate. A foundation for the proposed system was created, including hardware requirements, initial software stack choices, testing techniques and tools.

Future work will involve the steps mentioned in the methodology: gathering data, annotating, training a lightweight model, and evaluating its performance both offline and on-track in comparison to the pre-existing LiDAR system. This research aims to improve Monash Motorsport's technical knowledge on cameras and computer vision, provide an alternative vision system for future iterations of MMS vehicle design, and contribute to the discourse surrounding camera vision in autonomous vehicles.

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